



Politecnico di Milano

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New Technologies for Finance

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**Early-Warning System for Risk-Off Regimes
with Domain-Specific Allocation Routing**

Unsupervised Anomaly Detection for Regime-Aware Portfolio Allocation

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Problem & Objective

Traditional EWS treat risk-off as binary (equity vs. cash), ignoring crisis heterogeneity: in 2008 the USD strengthened (dollar funding squeeze); in inflationary regimes, gold outperforms. We build an EWS that answers *when* to go risk-off (ensemble anomaly detection) and *where* to allocate (macro-based routing: Cash USD / Gold / MBS). The strategy deploys **1.5× equity leverage** in calm periods and relies on early detection to avoid drawdowns, maximizing the Sharpe ratio.

Data & Features

Bloomberg weekly data, 2000–2021, 1,111 observations, 43 features (7 MSCI equity, 3 FX, 4 commodities, 8 bond TR, 18 yields, VIX, ECSURPUS). Target $Y=1$: risk-off week (21.3% prevalence). No missing values. Proxies: MSCI World→MXUS; Global Agg→LUACTRUU; credit spreads as log price-index ratios.

Features are made stationary (log-returns for prices, first differences for yields; 42/42 pass ADF at 5%). We construct 20 cross-asset spreads (term, credit, VRP, equity-bond rotation, gold-oil, JPY strength) and remove 7 collinear features ($|\rho|>0.9$), yielding **56 clean features**. Additionally, 14 routing triggers span three macro domains.

Validation: Walk-Forward CV with Embargo

Purged expanding walk-forward, 5 folds, **4-week embargo** between train and validation (eliminates leakage from Δ_{4w} lookbacks). Development ≤ 2018 ; test holdout 2019–2021 (sealed, contains COVID). Folds 1–2 have 53/64 positives (GFC, Euro Debt); folds 3–5 only 2/10/6 (low-vol 2013–18 regime). All metrics **weighted by n_{pos}** . Scaler fit only on each fold’s train; AE early-stopping on temporal sub-split of train only.

Anomaly Detection Models & Ensemble

Four novelty detectors trained on $Y=0$ weeks only:

- **MVG**: Mahalanobis distance, Ledoit-Wolf shrinkage (mandatory: 56 features, ~ 360 normals).
- **One-Class SVM (RBF)**: Grid $\nu \times \gamma$; best $\nu=0.05$, $\gamma=0.001$.
- **Autoencoder**: 56→24→12→6→12→24→56, dropout 0.15, early stopping on train sub-split.
- **Isolation Forest**: 200 trees, contamination=0.10.

Scores are percentile-mapped against train-normals distributions for comparability, then combined via soft mean voting. The **soft mean ensemble achieves $F1=0.650$** on the test holdout, outperforming the best single model (SVM, 0.615). Mean pairwise error correlation=0.66 confirms diverse error patterns.

	F1	AUC-ROC	AUC-PR	Prec	Rec	F2
MVG	0.516	0.886	0.770	1.000	0.348	0.400
SVM	0.615	0.851	0.732	0.750	0.522	0.556
IF	0.583	0.810	0.680	0.560	0.609	0.598
AE	0.516	0.873	0.764	1.000	0.348	0.400
Soft Mean Ens.	0.650	0.859	0.742	0.765	0.565	0.596

Routing Engine

When the ensemble signals risk-off, three domain sub-scores determine allocation. **USD sub-score**: mean of signed z-scores from funding stress (+1), DXY (+1), VRP (+1), Treasury flight (−1), US relative (+1)—dollar as safe haven during global, non-US stress (Avdjiev et al. 2019). **Gold sub-score**: real yield (−1), DXY (−1, opposite sign), JPY (+1), equity-bond ρ (+1), gold-oil (+1)—gold when stress is monetary/inflationary or USD is weak (Baur & Lucey 2010). **MBS**: binary rule-based ($VIX \in [20, 28]$, positive term spread, moderate drawdown), blocked if $VIX > 30$ or acute funding stress—defensive carry only. All μ, σ from development set only. Priority: USD→Gold→MBS→Cash (default). Thresholds optimized via Calmar-ratio grid search on CV folds.

Backtest Results (1.5× Leverage)

Tested on sealed 2019–2021 holdout with differentiated transaction costs (equity 5bps, cash 2bps, gold 8bps, MBS 20bps):

Allocation: 103 weeks levered equity (85.8%), 8 gold (6.7%), 8 cash (6.7%), 1 MBS (0.8%). During the COVID crash (Feb–Apr 2020), USD triggers fired first (funding stress, DXY rally, VRP spike); the system routed to cash and gold, with MBS correctly blocked ($VIX > 30$). During recovery (Apr–Dec 2020), allocations returned to 1.5× equity, capturing the rebound.

Limitations

	EWS Strategy	60/40	Buy & Hold
CAGR	35.2%	18.1%	24.2%
Sharpe	1.93	1.44	1.38
Max Drawdown	−12.1%	−21.5%	−27.7%
Calmar	2.92	0.84	0.87
Turnover	3.9/yr	—	—

Test holdout spans only 120 weeks with 9 regime switches—not yet statistically robust. Folds 3–5 are thin (2–10 positives). Proxies (MXUS≠MSCI World, log-ratios≠OAS) and weekly frequency limit applicability. The 2022 rate-shock regime is absent. At $1.5\times$ leverage, the 35% recall gap is a genuine concern.